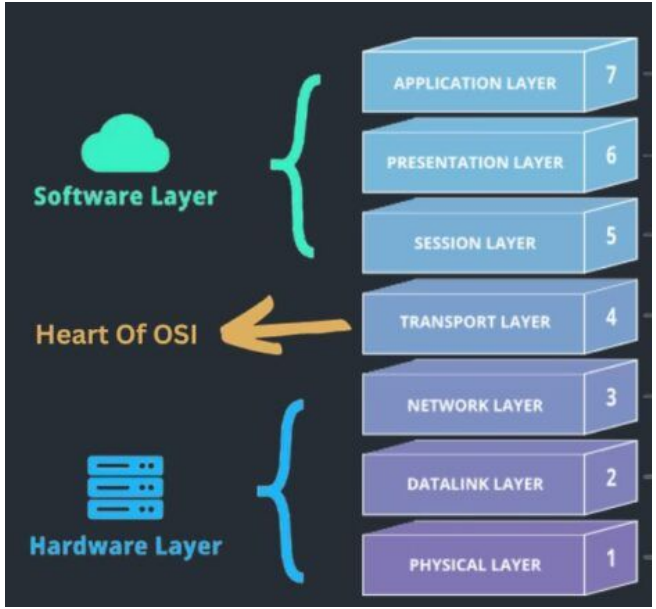
How do I enumerate

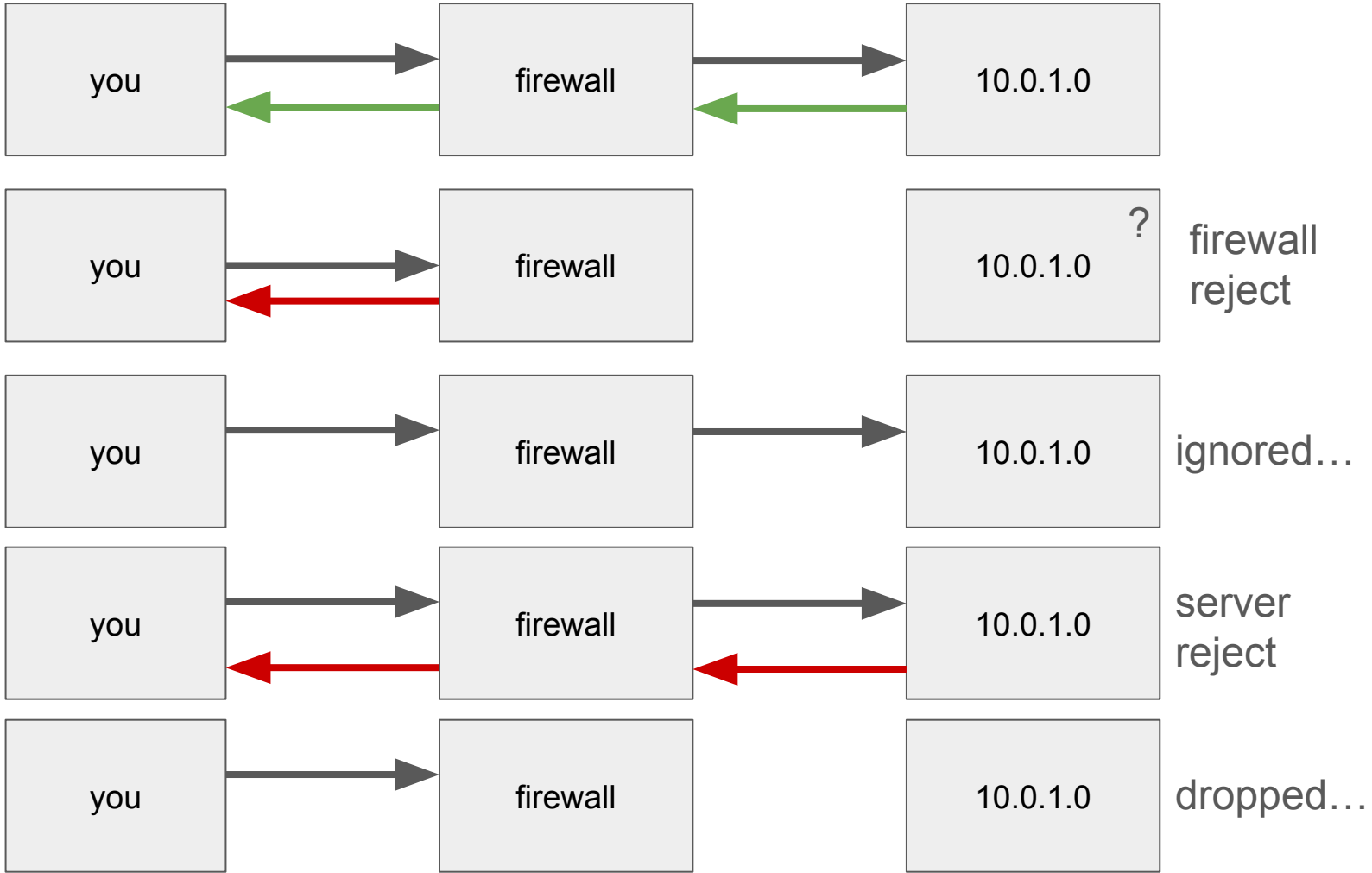
What is Enumeration

- We don’t know what we have (recall)
- Gathering information about the network and the boxes we work with.
- Attackers also don’t know what we have
- We need to figure out what we have before red team
- We also need to complete injects

Network basics

- UDP, TCP
  - user datagram protocol, transmission control protocol
  - UDP connectionless
  - TCP establishes connection, resending, etc
- QUIC, FTP, SSH, HTTP…
  - basically everything you usually work with
- ICMP
  - Internet Control Message Protocol, pings
- Port
  - some number, communication endpoint over UDP/TCP




you
firewall
10.0.1.0
you
firewall
10.0.1.0
?
firewall
reject
you
firewall
10.0.1.0
ignored…
you
firewall
10.0.1.0
server
reject
you
firewall
10.0.1.0
dropped…

nmap

- ip/port scanner
- can also run some scripts to interact with services, fetch banners, fingerprint OS/version, site content, smb shares, etc

- **nmap -sn 192.168.1.0/24**
- **sudo nmap -sS --top-ports 100 -T4 <target>**
- **nmap -sV -p 22,80,443 <target>** #detect service versions on ports
- **sudo nmap -O <target> #**
- **nmap --script vuln -p 80,443 <target>**
- **sudo nmap -A -p- -T4 -oN scan.txt <target>**

```
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-08-25 07:08 EDT
Nmap scan report for 192.168.36.135
Host is up (0.00097s latency).
Not shown: 471 closed tcp ports (reset)
PORT    STATE SERVICE     VERSION
21/tcp  open  ftp         vsftpd 2.3.4
22/tcp  open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp  open  telnet      Linux telnetd
25/tcp  open  smtp        Postfix smtpd
53/tcp  open  domain      ISC BIND 9.4.2
80/tcp  open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp open  rpcbind     2 (RPC #100000)
139/tcp open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
MAC Address: 08:00:27:CF:AD:DC (Oracle VirtualBox virtual NIC)
Service Info: Host:  metasploitable.localdomain; OSs: Unix, Linux; CPE: cpe
```

Enumerating Linux

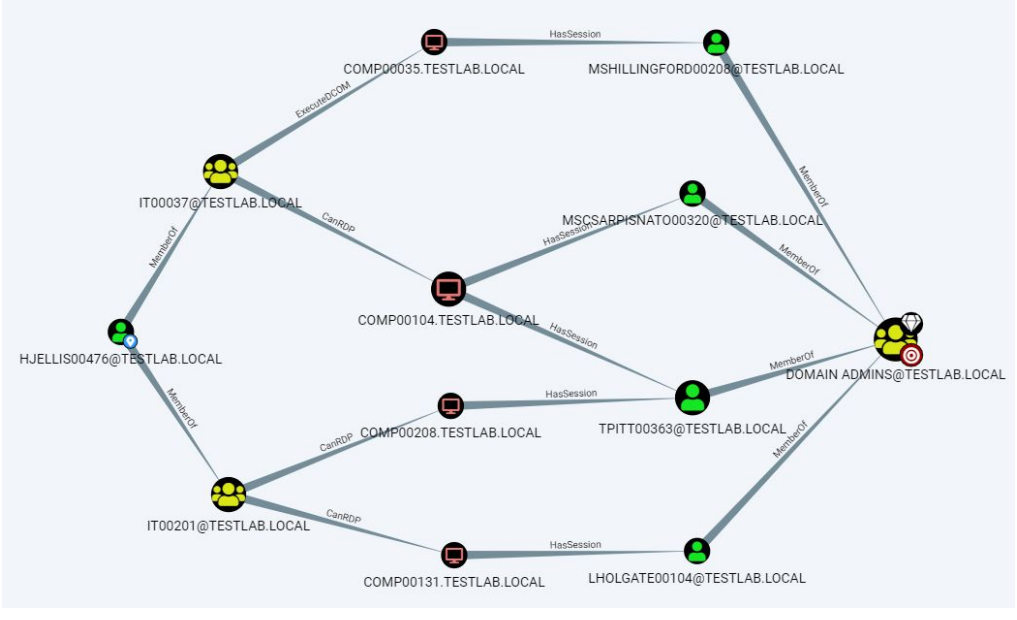
- Users, Groups, Permissions
  - Who has what permissions? Read/write/execute permissions on files? Do they need to have permissions? backupadmin on noisy-web
  - id → user id, groups, /etc/passwd, suid??
- Services
  - What services are running? On what ports? Under what user? What version?
  - ps aux
- OS version
- Connections
  - ss -anp
- Cron jobs
- Packages

Enumerating Windows

- Username and hostname
  - whoami
- Group memberships of users
  - whoami /groups
  - Get-LocalUser
  - Get-LocalGroup
  - Get-LocalGroupMember <group>
- Operating system, version, architecture
  - systeminfo
- Services, network
  - ipconfig /all
  - route print
  - netstat -ano # all active/listening tcp/udp connections
  - Service Binary Hijacking
- Installed 32x/64x applications
- Group Policies, Relationships, Password lockouts?!
- AD???

Automating Enumeration

- Many tools!
- linPEAS, winPEAS
  - “privilege escalation awesome script”
- Bloodhound
- nmap scripts



Q&A

1. I have all the info. What do I do with it?
    a. Change default creds always and note them down
    b. Find vulnerable versions and update them
    c. Disguised services (printer) - remove them
    d. Inspect malicious files and scripts
    e. Verify sus users and groups

POP QUIZ!

Our subnets:

- 10.0.1.0/24
- fd42:ccdc:1::/48

Connect to wifi!

- ssid: ows
- password: tryharder

Scoring VM addresses (out of scope):

- 10.0.1.225
- fd42:ccdc:1:8c01::1